
Instance-Level Costs for Nuanced Classifier Evaluation

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Abstract

Standard classification treats all errors equally, but in applications such as content moderation and medical screening, mistakes on clear-cut cases are more costly than errors on ambiguous ones. From a contextual bandit framework, we propose normalized excess cost (NEC), a metric that weighs classification errors by per-example costs and reduces to standard error rate when costs are uniform. Costs can derive from annotator vote margins, distance from decision thresholds, or confidence ratings. Across text, image, and tabular benchmarks, we find that NEC is often substantially lower than error rate—models with 5% error rate can achieve 1.8% NEC—revealing that most mistakes concentrate on ambiguous, low-cost examples. We also find that the evaluated methods—including cost-aware training, focal loss, and post-hoc thresholding—do not consistently improve NEC across datasets. Our framework provides a practical methodology for deriving and evaluating instance-level misclassification costs. Code and reproducibility artifacts are available at <https://github.com/KabirKang/instance-level-costs-paper-icml>.

classification cost reflecting its importance. This framing connects classification evaluation to decision theory: the cost-weighted error we minimize corresponds exactly to regret in a contextual bandit. These costs can be generated by annotator votes, distance to a threshold, or directly by annotator ratings.

Prior work on instance-dependent misclassification costs has largely focused on settings with objective, quantifiable costs, such as fraud detection or direct marketing (Höppner et al., 2022; Bahnsen et al., 2015; Vanderschueren et al., 2022). Unlike these works, we highlight the importance of modeling costs even when they are not as clearly definable. We show that the introduced *normalized excess cost* (NEC) is often significantly lower than the naive error, indicating that most errors are on borderline (low-cost) examples.

While we establish the construction of and evaluation with instance-level costs as providing a different, more nuanced view of errors, we also find mixed results for methods intended to improve NEC. In particular, we evaluate cost-aware strategies such as upsampling or upweighting points, training only on the highest-cost points, and directly regressing the signed margins, together with focal loss and post-hoc thresholding; no method consistently improves NEC across datasets. Thus, instance-level costs can substantially change evaluation even though the evaluated methods have dataset-dependent effects.

1. Introduction

Standard accuracy treats all mistakes as equally undesirable, but in practice this is often untrue. In content moderation and medical screening, misclassifying a clear-cut case is more serious than erring on a borderline example. This gap between uniform error counting and real-world consequences motivates evaluation and training methods that weight errors by their importance.

We propose a framework that assigns each example a mis-

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Contributions.

1. A general framework for deriving and using per-example misclassification costs in the normalized excess cost (NEC), equivalent to the normalized regret.
2. Experiments on image, text, tabular, and synthetic datasets showing NEC is significantly lower than standard misclassification error (equivalent to NEC with uniform costs).
3. An experimental comparison of training methods that leverage instance-level costs.

2. Related Work

2.1. Cost-Sensitive Classification

Cost-sensitive learning addresses classification problems where different types of errors incur different penalties. Elkan (2001) established theoretical foundations showing that optimal decision thresholds can be derived from class probabilities and misclassification costs, and that resampling training data is equivalent to adjusting these thresholds. Zadrozny et al. (2003) extended this framework to example-dependent costs through cost-proportionate weighting, where training examples are weighted by their associated misclassification costs. Langford & Beygelzimer (2005) provided a reduction from cost-sensitive multiclass classification to binary classification with formal regret guarantees, showing that instance-dependent cost vectors can be accommodated within their framework.

These methods assume costs are *externally specified*—typically by domain experts or derived from objective quantities. Bahnsen et al. (2015) exemplify this approach in fraud detection, where the cost of missing a fraudulent transaction (a false negative) equals the transaction amount, while a false positive incurs a fixed administrative cost of investigating the flagged transaction. Such financially-grounded costs are instance-dependent but objectively measurable. Our work addresses a complementary setting: when costs reflect *subjective* importance such as those derived from human disagreement, where the cost signal must be extracted from annotation patterns rather than external metadata. This shifts the problem from “how to train given costs” to “how to derive and validate costs from disagreement.”

A separate line of work questions whether instance weighting reliably changes what deep models learn. Byrd & Lipton (2019) examined importance weighting in deep learning, finding that its effect on over-parameterized models diminishes over successive epochs when the training data is separable. While their analysis does not consider disagreement-derived costs, it offers a prior explanation for why instance-level cost-weighting may fail to improve a converged model: on separable data, the weighting washes out. Our experiments complement theirs by examining cost-weighting derived from annotation margins and motivate, but do not establish, the hypothesis that effectiveness may depend on how cost-relevant information is represented in the features.

2.2. Human Label Variation and Task Ambiguity

A separate literature has examined settings where annotator disagreement reflects genuine ambiguity rather than noise. Dawid & Skene (1979) introduced the foundational EM-based approach for jointly estimating true labels and annotator reliability, treating disagreement as a nuisance to be marginalized. Plank et al. (2014) challenged this view,

demonstrating that inter-annotator disagreement in part-of-speech tagging is systematic and linguistically motivated rather than erroneous. This perspective—that disagreement carries signal—has gained substantial traction.

Pavlick & Kwiatkowski (2019) showed that disagreements in natural language inference persist even with additional context, reflecting inherent ambiguity in the task itself. Peterson et al. (2019) collected full label distributions for CIFAR-10 and demonstrated that training on soft labels improves robustness and out-of-distribution generalization. Nie et al. (2020) scaled this approach with ChaosNLI, collecting 100 annotations per example and revealing that models achieve near-perfect accuracy on high-agreement examples but perform at chance on low-agreement ones. Uma et al. (2021) surveyed methods for learning from disagreement, cataloging approaches ranging from soft-label training to multi-annotator modeling. Plank (2022) synthesized these developments in a position paper arguing that human label variation should be embraced throughout the ML pipeline.

Most recently, Kurniawan et al. (2025) proposed soft evaluation metrics based on fuzzy set theory, where soft accuracy measures overlap between predicted and human judgment distributions. Their extensive empirical study found that two simple strategies for preserving annotator disagreement during training—using the full label distribution from the votes (*soft labels*), or treating each annotator’s judgment as a separate training instance (*disaggregated annotations*)—outperform more complex approaches. However, soft metrics evaluate *calibration*—whether the model’s output distribution matches the human distribution—rather than *decision quality*. Our metric instead treats disagreement magnitude as misclassification cost: a model that correctly classifies high-consensus examples while erring on ambiguous ones incurs lower expected cost than the reverse, even if both models have identical accuracy.

2.3. Predicting Disagreement

Several methods leverage predicted disagreement for purposes other than evaluation. Raghu et al. (2019) train models to predict which medical cases will elicit expert disagreement, using these predictions to route difficult cases for second opinions. Their key finding—that directly predicting uncertainty outperforms deriving it from classifier confidence—demonstrates that disagreement is feature-predictable. We build on this insight but pursue a different goal: rather than routing examples, we use disagreement magnitude to weight evaluation, asking whether models that minimize cost-weighted error differ from those minimizing unweighted error (i.e., standard misclassification error).

2.4. This Work’s Position

Our work bridges cost-sensitive classification and ambiguity in classification, including human label variation. From the former, we adopt the framework of instance-dependent misclassification costs and expected-cost minimization. From the latter, we take the insight that examples have varying levels of ambiguity and that annotator disagreement reflects meaningful signal about example ambiguity. The gap we address is that cost-sensitive methods assume costs are given, while disagreement methods use variation for training or calibration rather than cost-based evaluation. We propose deriving per-example costs when they aren’t easily quantifiable, and evaluating models by normalized excess cost, connecting to decision-theoretic notions of regret.

3. Cost-Sensitive Classification

3.1. Motivation

Standard accuracy assumes every mistake is equally costly, yet for many practical tasks, some errors matter more than others. The challenge is that “importance” is often subjective: unlike fraud amounts or medical expenditures, there is no external ledger recording the cost of misclassifying a borderline content moderation case. We argue that even *approximate* costs are more informative than the implicit assumption of uniform costs. If annotators split 6–4 on whether a comment is toxic, either classification is defensible; if they agree 10–0, misclassification is a clear failure. Normalized Excess Cost (NEC) operationalizes this intuition without requiring that costs be precisely calibrated.

3.2. Decision-Theoretic Regret

We consider a binary action setting (actions $+1$ and -1) where each example consists of a triple $(x, r^{(-1)}, r^{(+1)})$. $x \in \mathcal{X}$ is the input, $r^{(-1)} \in \mathbb{R}$ is the reward if the -1 action is chosen and $r^{(+1)} \in \mathbb{R}$ is the reward if the $+1$ action is chosen. Our goal is to find a policy $\pi : \mathcal{X} \rightarrow \{-1, +1\}$ which minimizes the regret $R(\pi) = \mathbb{E}[\max_{a \in \{-1, +1\}} r^{(a)}] - \mathbb{E}[r^{(\pi(x))}]$.

Specific to binary actions, we can add additional notation to simplify the problem. For a given example, define $\Delta = r^{(+)} - r^{(-)}$. Then the regret is simply $R(\pi) = \mathbb{E}_{x, \Delta} [|\Delta| \cdot \mathbb{1}[\text{sign}(\Delta) \neq \pi(x)]]$. In this form, we see a close similarity to binary classification with a label $y = \text{sign}(\Delta)$ and an instance-level misclassification cost of $|\Delta|$. The metric of standard accuracy has $\Delta \in \{-1, 1\}$ and the metric of cost-sensitive classification (with false positive and false negative costs a and b , respectively) has $\Delta \in \{-a, b\}$.

3.3. Classification Evaluation

In order to produce an interpretable metric with the same scale as the misclassification error, we scale the regret by the maximum possible regret. For a classifier’s predictions $\hat{y}_i$, we refer to this quantity as the normalized excess cost (NEC) where “excess cost” is synonymous with “regret”,

$$\text{NEC} = \frac{\sum_{i=1}^n |\Delta_i| \cdot \mathbb{1}[\hat{y}_i \neq \text{sign}(\Delta_i)]}{\sum_{i=1}^n |\Delta_i|}. \quad (1)$$

When $|\Delta_i| = 1$ for all i , NEC reduces to error rate, making the two metrics directly comparable.

$$\text{Error Rate} = \frac{1}{n} \sum_{i=1}^n \mathbb{1}[\hat{y}_i \neq \text{sign}(\Delta_i)]. \quad (2)$$

A method which randomly guesses receives 0.5 NEC and 0.5 error rate on average.

The ratio of the Error Rate to the NEC characterizes how much better a model performs on high-cost examples relative to low-cost ones. When this ratio exceeds 1, the model’s errors concentrate on ambiguous cases, which is a desirable property that error rate alone cannot detect. NEC measures the fraction of total cost incurred by misclassification. Lower NEC indicates better alignment with task-specific priorities: a model can achieve low NEC by correctly classifying high-cost examples even if it errs on many ambiguous, low-cost cases.

Our formulation provides a decision-theoretic grounding for instance-level classification costs. From a decision-theoretic perspective, high $|\Delta|$ indicates decisions with large reward gaps where correct classification matters greatly, while low $|\Delta|$ indicates near-indifference where either action yields similar reward. This interpretation is agnostic to cost provenance and holds whether costs derive from annotator disagreement, clinical thresholds, or any other source.

3.4. Cost Derivation

Our framework applies whenever per-example costs are available. We describe three natural sources of costs.

Multi-annotator labels. When multiple annotators label each example, their agreement provides a natural cost signal. Let $n_{i,\text{yes}}$ and $n_{i,\text{no}}$ denote the number of positive and negative votes for example i . The Laplace-smoothed estimate of the yes frequency is $(n_{i,\text{yes}} + 1)/(n_{i,\text{yes}} + n_{i,\text{no}} + 2)$. We define the reward difference as the log-odds of the smoothed yes vote estimate:

$$\Delta_i = \log \frac{n_{i,\text{yes}} + 1}{n_{i,\text{no}} + 1}, \quad (3)$$

Note that Laplace smoothing ensures finite values when annotators unanimously agree. Log-odds is the natural

scale for binary outcomes: it is symmetric around zero, unbounded, and $|\Delta|$ is monotonically related to distance from maximum uncertainty. High $|\Delta|$ indicates strong annotator consensus where misclassification is costly; $|\Delta| \approx 0$ indicates ambiguity where either label is defensible.

Threshold- and rating-based labels. When binary labels derive from thresholding a continuous measurement, the distance to the threshold provides a natural cost. Let z_i be the continuous value and τ the decision threshold. We define:

$$\Delta_i = z_i - \tau, \quad (4)$$

A patient whose measurement is far above or below the threshold represents a clear-cut case where misclassification is serious; a patient near the threshold is a borderline case where the label itself is less certain. Ordinal ambiguity or confidence ratings are a special case of this construction: subtracting the midpoint of a 5-point or 7-point scale yields a signed margin, as in the iNaturalist experiment.

3.5. Training with Costs

In addition to standard cross-entropy training, we compare several approaches that use the costs $|\Delta_i|$. Intuitively, cost-weighting encourages the model to prioritize high-cost examples during training. Since our evaluation metric weights errors by cost, aligning the training objective with the evaluation metric should improve performance—a principle analogous to importance weighting in domain adaptation (Shimodaira, 2000).

Standard training. The standard approach minimizes binary cross-entropy loss on the label $\text{sign}(\Delta_i)$. Let $\hat{p}_i \in [0, 1]$ denote the model’s predicted probability of the positive class +1, and then at inference, we predict $\hat{y}_i = +1$ if $\hat{p}_i \geq 0.5$, and $\hat{y}_i = -1$ otherwise. The standard training loss is,

$$\mathcal{L}_{\text{cls}} = - \sum_{i=1}^n [\mathbf{1}[\Delta_i \geq 0] \log \hat{p}_i + \mathbf{1}[\Delta_i < 0] \log(1 - \hat{p}_i)]. \quad (5)$$

Cost-weighted training. To emphasize high-cost examples during training, we weight each example’s loss by its cost:

$$\mathcal{L}_{\text{wt}} = - \sum_{i=1}^n |\Delta_i| \cdot [\mathbf{1}[\Delta_i \geq 0] \log \hat{p}_i + \mathbf{1}[\Delta_i < 0] \log(1 - \hat{p}_i)]. \quad (6)$$

Cost-weighting modifies only the training objective; at inference, we use the same decision rule of predicting +1 if and only if $\hat{p}_i \geq 0.5$.

Sampling strategies. Rather than modifying loss weights, we can alter the training distribution. We evaluate two approaches applied per-class to preserve class balance: (1) *probability-proportional upsampling*, which resamples with replacement where selection probability is proportional to $|\Delta_i|$, maintaining the original dataset size while increasing the effective influence of high-cost examples; and (2) *top-k% filtering*, which retains only examples with $|\Delta_i|$ at or above the $(100-k)$ th percentile within each class. We test $k \in \{30, 50, 70\}$, corresponding to keeping the highest-cost 30%, 50%, or 70% of examples respectively.

Δ -regression. Rather than predicting a binary label, we train a regression model to predict the signed margin Δ_i directly using squared error loss:

$$\mathcal{L}_{\text{reg}} = \sum_{i=1}^n (\hat{\Delta}(x_i) - \Delta_i)^2. \quad (7)$$

At inference, we classify by thresholding at zero: $\hat{y} = \text{sign}(\hat{\Delta}(x))$. Regression yields a continuous $\hat{\Delta}$ that can be used for ranking or downstream selective prediction, analogous to a calibrated confidence estimate.

4. Experiments

4.1. Experimental Setting

Datasets. We evaluate on four datasets spanning text, image, and tabular domains (Table 1).

Jigsaw contains online comments from the Civil Comments archive, annotated for toxicity by multiple crowd annotators (Borkan et al., 2019) and we use the Laplace-smoothed annotation log-odds as Δ_i , as described in Section 3.4.

Turkey contains images of turkeys from barn surveillance cameras, labeled by multiple annotators for visible injuries (Volkman et al., 2021; Schmarje et al., 2022), and we use the Laplace-smoothed annotation log-odds as Δ_i .

NHANES contains tabular health data from the CDC’s National Health and Nutrition Examination Survey (2013–2014) (National Center for Health Statistics (NCHS), 2014). The task is hypertension classification using the 2017 ACC/AHA guideline threshold of 130 mmHg systolic blood pressure, with $\Delta_i = \text{SBP}_i - 130$. Features include age, gender, race/ethnicity, and BMI.

iNaturalist contains plant images sampled from the iNaturalist API (iNaturalist, 2024), labeled by Gemini 2.5-pro (Google DeepMind, 2025) on a 7-point confidence scale from “indisputably wild” (1) to “indisputably cultivated” (7). We use $\Delta_i = 4 - \text{rating}_i$, demonstrating cost derivation from a single annotator’s confidence rather than multi-annotator disagreement.

To understand the performance of methods in an idealized

setting, we also construct a *Synthetic* dataset where the signed margin is a linear function of the input features plus Gaussian noise. We use a unit weight vector $w \in \mathbb{R}^d$ (with $d = 50$) and isotropic Gaussian features $x_i \sim \mathcal{N}(0, I_d)$, and set

$$\Delta_i = w^\top x_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 0.1^2), \quad (8)$$

Since the signal term $w^\top x_i$ has unit variance while the noise standard deviation is only 0.1, $|\Delta_i|$ is the magnitude of this noisy linear margin and is almost entirely determined by the features—an idealized case in which methods that use Δ have access to a strong feature-derived signal.

Dataset	N	Input	Target	Cost source
Jigsaw	1.8M	Text	Toxic	Human votes
Turkey	8,040	Image	Injured	Human votes
iNaturalist	9,956	Image	Wild	LLM rating
NHANES	7,455	Tabular	Hypertensive	Blood Pressure

Table 1. Datasets used in experiments. Costs derive from multi-annotator disagreement, distance to a decision threshold, or confidence ratings.

Models. We use standard architectures appropriate to each domain: TF-IDF features and frozen RoBERTa (Liu et al., 2019) embeddings with logistic regression for Jigsaw, frozen ImageNet-pretrained ResNet-50 (He et al., 2016) embeddings with logistic regression for Turkey and iNaturalist, and HistGradientBoosting for NHANES tabular features. We also evaluate end-to-end fine-tuning of RoBERTa for Jigsaw and ImageNet-pretrained ResNet-50 for Turkey and iNaturalist.

Implementation details. All linear models use L2 regularization with default hyperparameters. For the real datasets, we use an 80/10/10 train/validation/test split stratified on the label $\text{sign}(\Delta)$; Synthetic uses 900 training, 100 validation, and 10,000 test examples. We report results over 10 random seeds with mean ± 1.96 times the standard error (corresponding to a 95% Gaussian confidence interval).

4.2. NEC vs. Error Rate

We first compare NEC against error rate across datasets and models, then examine how the two metrics scale with training-set size, how NEC relates to other evaluation metrics, and how robust it is to the choice of Δ transform. Section 4.3 then evaluates methods intended to improve NEC.

NEC vs error rate. Figure 1 presents the main NEC-versus-error comparison, and Appendix Table 3 reports the exact values. The key finding is that *error rate and NEC can diverge substantially*, revealing different pictures of model performance.

On Jigsaw with TF-IDF, the model achieves 5.34% error rate but only 1.76% NEC—error rate is more than three times higher. This indicates that most errors occur on low-cost examples where annotators disagreed, while the model performs well on high-cost examples with strong annotator consensus.

The gap between error rate and NEC varies by dataset. On Turkey, error rate (6.68%) is $1.6\times$ NEC (4.16%). On NHANES, error rate (21.68%) is $1.5\times$ NEC (14.83%). On iNaturalist, the ratio is smaller: error rate (11.73%) is only $1.1\times$ NEC (10.39%). The Synthetic control shows the largest gap by construction (4.55% error rate vs. 0.50% NEC, a $9\times$ ratio), reflecting that signed Δ is a linear function of the features plus Gaussian noise, $|\Delta|$ is the magnitude of that noisy linear margin, and most errors land on low-cost examples near the decision boundary. Under fine-tuning, both metrics improve substantially—on Jigsaw, RoBERTa-FT reduces NEC from 3.29% to 1.78% (46% relative) and error rate from 7.56% to 4.73%; on Turkey, ResNet-FT reduces NEC from 4.16% to 2.27% and error rate from 6.68% to 4.75%—while the same per-dataset ordering is preserved (error-to-NEC ratio is $2.7\times$ on Jigsaw RoBERTa-FT, $2.1\times$ on Turkey, $1.2\times$ on iNaturalist), showing that the NEC/error gap persists under fine-tuning while its magnitude varies across model regimes. The Δ distributions for each dataset are shown in Figure 2.

From a practitioner’s standpoint, the gap means a system reported at 5% error rate can have substantially better cost-weighted performance than accuracy suggests—on Jigsaw, a 5%-error model is also a 1.8%-NEC model, since most errors land on the low-cost ambiguous cases that humans disagree on. The size of the gap also informs the task itself: a small gap (as on iNaturalist) indicates the model performs similarly on high cost and low cost examples, while a large gap (as on Jigsaw) signals many ambiguous examples where the model errs.

Sample size scaling. Figure 3 shows the error rate and NEC as a function of the nominal pre-split sample size N on Jigsaw, varying N from 1,000 to 100,000. The data are sampled to N before the 80/10/10 train/validation/test split, so the training, validation, and test sets all increase with N ; both metrics improve overall as nominal N increases.

Comparison to soft and class-aware metrics. Brier score (Brier, 1950) and log-loss evaluate probabilistic predictions, while Balanced Accuracy (Brodersen et al., 2010) and F_2 (van Rijsbergen, 1979) emphasize class-level performance. To compare these metrics with NEC, we compute Spearman correlations across the six shared classification strategies—Standard CE, probability-proportional upsampling, $|\Delta|$ -weighted CE, and top-30%, top-50%, and top-70% filtering—in each of five Figure 4 panels with complete

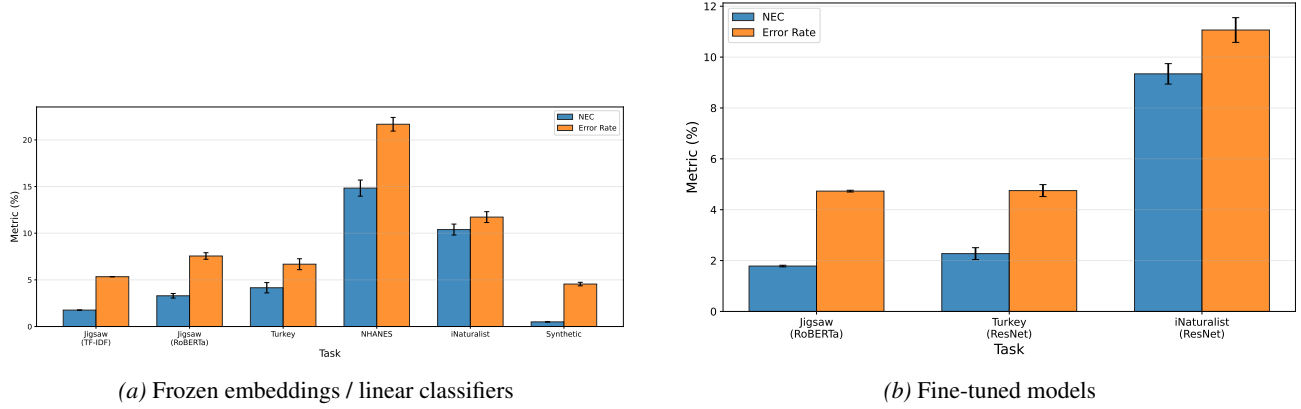


Figure 1. NEC vs Error Rate across datasets. Blue bars show NEC (cost-weighted error); orange bars show standard error rate. Fine-tuning improves both metrics while preserving the NEC/error-rate gap.

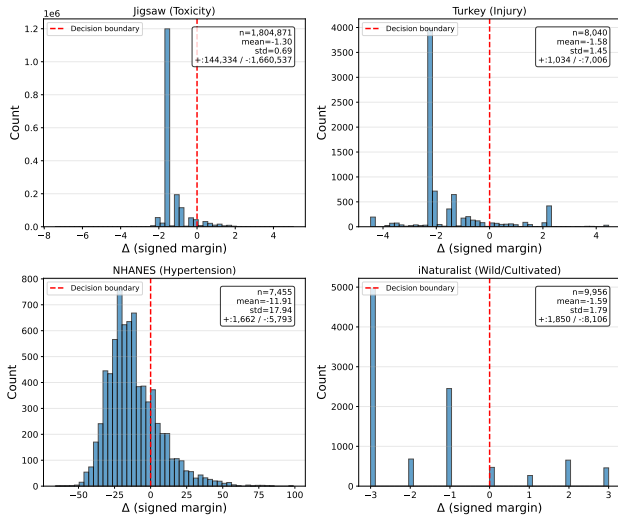


Figure 2. Distribution of signed Δ across datasets. All histograms are oriented so the minority class is on the right (positive) side. The red dashed line marks the decision boundary at $\Delta = 0$.

per-example predictions (Appendix A.2). NEC and error rank the methods similarly but not identically ($\rho = 0.71$ –1.00). Correlations with Brier, log-loss, $1 - \text{BalAcc}$, and $1 - F_2$ vary substantially across panels, including changes in sign, so no single correlation summarizes their relationship with NEC.

Sensitivity to the Δ transform. The absolute value of NEC depends on the choice of Δ transform. For multi-annotator data (Jigsaw, Turkey), we compare the smoothed log-odds used throughout (default), the normalized vote margin $\Delta = (n_{\text{yes}} - n_{\text{no}})/n_{\text{total}}$, and a Laplace-smoothed entropy cost, where $p = \frac{n_{\text{yes}} + 1}{n_{\text{yes}} + n_{\text{no}} + 2}$ and $|\Delta|_{\text{entropy}} = 1 - \frac{H(p)}{\log 2}$. For real-valued data (NHANES, iNaturalist), we compare the margin $\Delta = z - \tau$ (default) to using the squared costs $|\Delta| = |z - \tau|^2$. Table 2 reports the Standard training

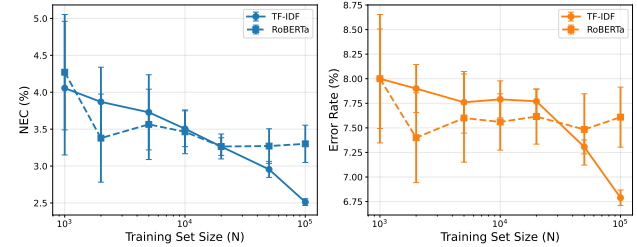


Figure 3. NEC and Error Rate as a function of nominal pre-split sample size N on Jigsaw. Both metrics improve overall as nominal N increases.

NEC under each transform. The transform changes absolute NEC and can also change method rankings; the full results are reported in Appendix B.

Table 2. NEC (%) of Standard training under alternative Δ transforms. Default transform is smoothed log-odds for multi-annotator data and raw margin $|\Delta|$ for real-valued data.

Transform	Jigsaw	Turkey	NHANES	iNat
Default	1.76	4.16	14.83	10.39
Vote margin	1.64	4.60	—	—
Smoothed entropy	1.25	3.69	—	—
Squared margin	—	—	13.47	9.27

Appendix C compares NEC and error along confidence-ranked coverage curves for the Standard models; both generally decrease as coverage decreases, and mean NEC is no greater than mean error at any plotted coverage.

4.3. Cost-Sensitive Training Methods

4.3.1. LOGISTIC REGRESSION AND FIXED-REPRESENTATION MODELS

The methods introduced in Section 3.5 are Standard CE, probability-proportional upsampling, $|\Delta|$ -weighted CE, top-

30%, top-50%, and top-70% filtering, and Δ -regression. We additionally evaluate calibrated $|\Delta|$ -reweighting, Costing following [Zadrozny et al. \(2003\)](#), and Calibrated Elkan-style thresholding following [Elkan \(2001\)](#). Figure 4 compares these methods for logistic-regression, fixed-representation, and classical base-model settings. Missing bars indicate methods not evaluated for that dataset.

No method consistently improves NEC across datasets, and effects vary by dataset. Δ -regression gives the clearest gain on Synthetic (0.50% $\rightarrow$ 0.26%) but is mixed on the real datasets. Calibrated $|\Delta|$ -reweighting, Costing, and Calibrated Elkan likewise provide no consistent cross-dataset improvement.

4.3.2. FINE-TUNING

Figure 5 compares $|\Delta|$ -weighting and focal loss ([Lin et al., 2017](#)), each against its corresponding cross-entropy control; focal $\gamma = 0$ is the control for the focal-loss comparison.

In the weighting comparison, $|\Delta|$ -weighting is essentially unchanged on Jigsaw, is not distinguishable from the Standard baseline on Turkey, and improves NEC on iNaturalist. Paired comparisons support small NEC improvements from focal loss on Jigsaw for all three tested nonzero γ values; the largest occurs at $\gamma = 3$, from 1.775% to 1.738% (mean paired difference -0.037 percentage points; 95% CI $[-0.055, -0.018]$). Focal loss is not distinguishable from its baseline on Turkey and iNaturalist. No method consistently improves NEC across all three fine-tuned datasets.

Full numerical results are available in the public code repository.

5. Discussion

5.1. Interpreting the Mixed Results

Across the evaluated methods, effects vary by dataset and method, and no approach consistently improves NEC across datasets.

One possible explanation for the mixed cost-aware training results is that these methods are more useful when the features contain information about the cost magnitude. The Synthetic experiment and the iNaturalist direction change motivate, rather than establish, this hypothesis.

5.2. Limitations

Our cost derivation assumes that annotator disagreement or distance to threshold correlates with true importance. This assumption need not hold exactly: NEC remains more informative than error rate as long as costs are *ordinally* correct (high- $|\Delta|$ examples genuinely matter more than low- $|\Delta|$ ones). When external costs exist (e.g., user impact,

legal liability), these can replace or augment disagreement-derived costs within the same framework.

The iNaturalist dataset relies on LLM-generated confidence ratings rather than human annotator votes. To understand this cost signal, we compared LLM confidence against held-out human judgments on Jigsaw and Turkey samples. Agreement on the binary label is moderate (72.6% on $n=500$ Jigsaw items, 68.0% on $n=300$ Turkey items), but the correlation between LLM-derived $|\Delta|$ and human-derived $|\Delta|$ is weak on Jigsaw (Spearman $\rho = 0.087$, $p = 0.052$) and moderate on Turkey ($\rho = 0.249$, $p = 1.24 \times 10^{-5}$). LLM confidence captures only some of the signal captured by multi-annotator disagreement. The iNaturalist experiment illustrates how a single-rater scale can be incorporated into the framework, but LLM-derived costs require domain validation and should not be treated as a substitute for human-derived disagreement costs. Our conclusions are limited to the evaluated methods and configurations.

Impact Statement

This paper presents methods for evaluating classifiers using instance-level misclassification costs derived from annotator disagreement, decision thresholds, or confidence ratings. We highlight several societal considerations.

Potential benefits. Our framework encourages practitioners to recognize that not all classification errors are equally consequential. In content moderation, medical screening, and safety-critical systems, this perspective could lead to more nuanced evaluation that better reflects real-world priorities and rewards systems that correctly handle clear-cut cases even if they struggle with genuinely ambiguous ones.

Potential risks. Cost derivation from annotator disagreement assumes that consensus reflects importance, which may not hold universally. In some domains, unanimous agreement could reflect systematic bias rather than ground truth clarity. Additionally, framing ambiguous cases as “low-cost” could be misused to justify poor performance on examples that are difficult precisely because they affect marginalized groups whose perspectives are underrepresented among annotators. We recommend two mitigations when applying this framework in deployed settings. First, the annotator pool should be diverse along dimensions relevant to the task (demographics, lived experience, domain expertise), and the pool composition should be reported alongside any cost-weighted evaluation. Second, before treating $|\Delta|$ as a cost signal, practitioners should check whether $|\Delta|$ correlates with sensitive attributes or protected-group membership in the data; systematic correlation between low agreement and a particular subgroup is a signal that “ambiguity” may be tracking bias rather than genuine task difficulty.

Instance-Level Costs for Nuanced Classifier Evaluation

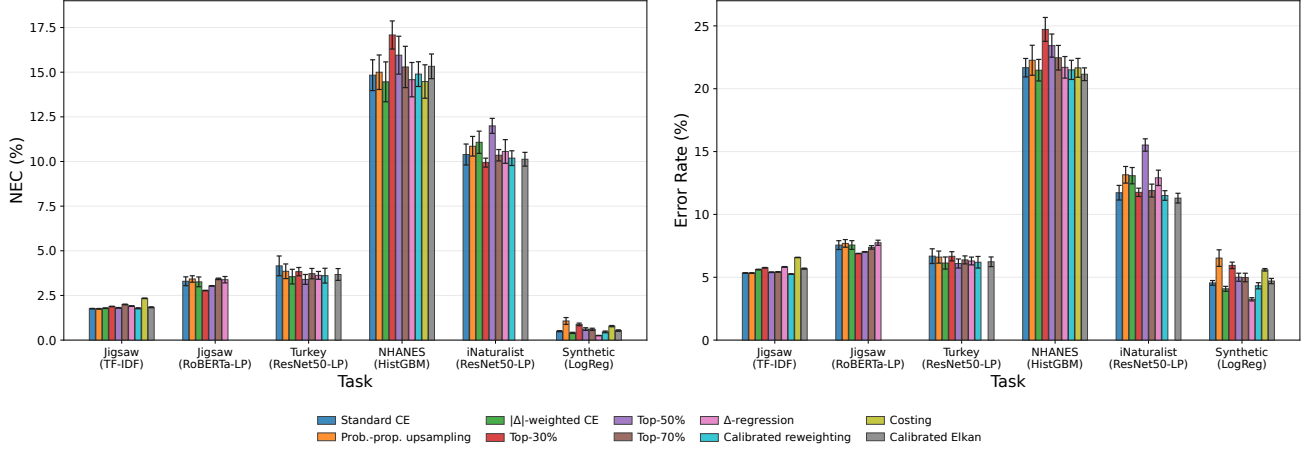


Figure 4. NEC and Error Rate for fixed-representation and base-model methods. Bars show means over ten seeds and error bars show 95% confidence intervals. Missing bars indicate combinations that were not evaluated. No method consistently improves NEC across datasets.

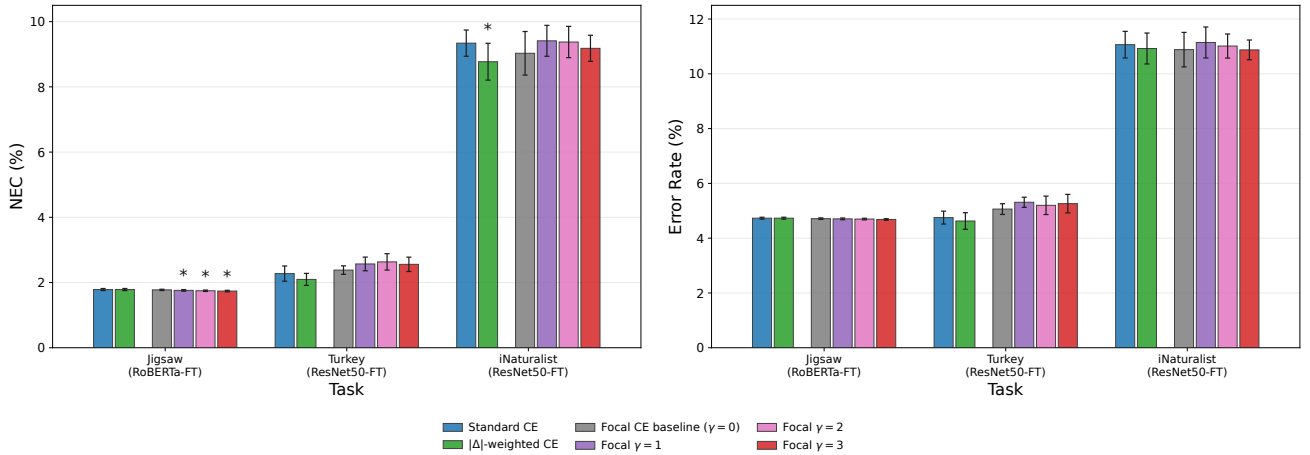


Figure 5. NEC and Error Rate for fine-tuned models. Methods are grouped into weighting and focal-loss comparisons, each with its corresponding cross-entropy control; focal $\gamma = 0$ is the control for focal loss. Bars show means over ten seeds with 95% confidence intervals. Asterisks in the NEC panel mark method-control comparisons whose paired 95% confidence intervals for the NEC difference exclude zero.

Scope. Our experiments use publicly available datasets for content moderation, medical diagnosis, and species classification. For the iNaturalist dataset, we used LLM-generated annotations to derive costs rather than human annotators. We do not deploy systems in practice. The primary contribution is methodological, by offering a different lens for evaluation, rather than building deployed applications.

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A. Supplemental Numerical Results

A.1. Detailed Baseline Results

Table 3. Main results: NEC and Error Rate (%) for Standard cross-entropy training across datasets. Models with “-LP” use frozen pretrained embeddings with a linear classifier; “-FT” indicates end-to-end fine-tuning. Mean $\pm$ 95% CI over 10 seeds. Lower is better.

Task	Model	NEC	Error
Jigsaw	TF-IDF	1.8 $\pm$ 0.0	5.3 $\pm$ 0.0
Jigsaw	RoBERTa-LP	3.3 $\pm$ 0.2	7.6 $\pm$ 0.4
Jigsaw	RoBERTa-FT	1.8 $\pm$ 0.0	4.7 $\pm$ 0.0
Turkey	ResNet50-LP	4.2 $\pm$ 0.6	6.7 $\pm$ 0.6
Turkey	ResNet50-FT	2.3 $\pm$ 0.2	4.8 $\pm$ 0.2
NHANES	HistGBM	14.8 $\pm$ 0.9	21.7 $\pm$ 0.7
iNaturalist	ResNet50-LP	10.4 $\pm$ 0.6	11.7 $\pm$ 0.6
iNaturalist	ResNet50-FT	9.3 $\pm$ 0.4	11.1 $\pm$ 0.5
Synthetic	LogReg	0.5 $\pm$ 0.0	4.6 $\pm$ 0.2

A.2. Metric-Ranking Comparisons

For each row of Table 4, we average each metric over the ten canonical seeds for the six shared classification strategies (Standard CE, probability-proportional upsampling, $|\Delta|$ -weighted CE, and top-30%, top-50%, and top-70% filtering), and then rank those strategies within the corresponding Figure 4 panel. The table reports Spearman ρ . We orient $1 - \text{BalAcc}$ and $1 - F_2$ so lower is better, like NEC. Table 4 includes the five Figure 4 panels for which the required per-example predictions are available.

Table 4. Spearman rank correlations between NEC and other metrics across six classification strategies within each Figure 4 panel. Each strategy metric is averaged over ten canonical seeds before ranking.

Panel	Error	Brier	Log-loss	$1 - \text{BalAcc}$	$1 - F_2$
Jigsaw TF-IDF	0.71	0.54	0.26	0.09	0.09
Turkey ResNet50-LP	0.94	0.14	-0.49	-0.26	0.83
NHANES HistGBM	1.00	0.89	0.89	-0.94	-0.94
iNaturalist ResNet50-LP	0.77	0.77	0.31	-1.00	0.94
Synthetic LogReg	1.00	0.89	0.77	1.00	0.94

B. Δ -Transform Sensitivity

The NEC formulation depends on the choice of Δ transform from raw annotator agreement signals to a per-example cost. Section 4.2 reports the absolute NEC values under each transform; this appendix presents the supporting tables, including rank correlations between configuration orderings.

We evaluate Δ transforms in two families. For **multi-annotator data** (Jigsaw, Turkey), where Δ derives from positive vs. negative vote counts, we compare three transforms: smoothed log-odds $\Delta_i = \log \frac{n_{i,\text{yes}}+1}{n_{i,\text{no}}+1}$ (the default used throughout this paper), normalized vote margin $|n_{i,\text{yes}} - n_{i,\text{no}}|/n_{i,\text{total}}$, and Laplace-smoothed entropy with $p = \frac{n_{i,\text{yes}}+1}{n_{i,\text{yes}}+n_{i,\text{no}}+2}$ and $|\Delta|_{\text{entropy}} = 1 - \frac{H(p)}{\log 2}$. For **real-valued data** (NHANES, iNaturalist), where Δ derives from distance to a threshold or annotator rating, we compare two transforms: the raw margin $|\Delta|$ (default) and the squared margin Δ^2 . Entropy is not applicable to real-valued datasets, as they have no vote counts.

For each transform, we report two quantities: absolute NEC of the Standard CE classifier under that transform (averaged over 10 random seeds on the same full-dataset runs as Table 3), and Spearman rank correlation between the six-strategy ranking under the default transform and the ranking under the alternative after averaging each strategy over ten canonical seeds. Tables 5 and 6 report these for multi-annotator and real-valued datasets, respectively.

Table 5. Δ -transform sensitivity on multi-annotator datasets. Absolute NEC (%) is for Standard CE under each transform (matching Table 3 at the default). Spearman ρ is the rank correlation between the configuration ordering under the default transform and the alternative.

Transform	NEC (%)		Rank correlation ρ	
	Jigsaw	Turkey	Jigsaw	Turkey
Default (log-odds)	1.76	4.16	—	—
Normalized margin	1.64	4.60	0.83	1.00
Smoothed entropy	1.25	3.69	0.94	1.00

Table 6. Δ -transform sensitivity on real-valued datasets. Absolute NEC (%) is for Standard CE under each transform (matching Table 3 at the default). Spearman ρ is the rank correlation between the configuration ordering under the default transform and the alternative. Entropy is not applicable.

Transform	NEC (%)		Rank correlation ρ	
	NHANES	iNaturalist	NHANES	iNaturalist
Default ($ \Delta $)	14.83	10.39	—	—
Squared margin	13.47	9.27	0.49	1.00

Ranking sensitivity is dataset-dependent. The alternative transforms preserve the six-method ordering on Turkey and

iNaturalist and remain strongly correlated with the default on Jigsaw ($\rho = 0.83$ for normalized margin and 0.94 for smoothed entropy). NHANES is more sensitive to squaring the margin ($\rho = 0.49$). Absolute NEC values also shift with the transform.

C. Selective Classification: NEC Versus Error

Selective classification asks how model performance changes when the system is allowed to abstain from low-confidence predictions (Raghu et al., 2019). At each coverage level $c \in [0, 1]$, we restrict evaluation to the top- c fraction of test examples ranked by predictive confidence ($\max(\hat{p}, 1 - \hat{p})$). We compute NEC and Error Rate on the same retained subset at every coverage.

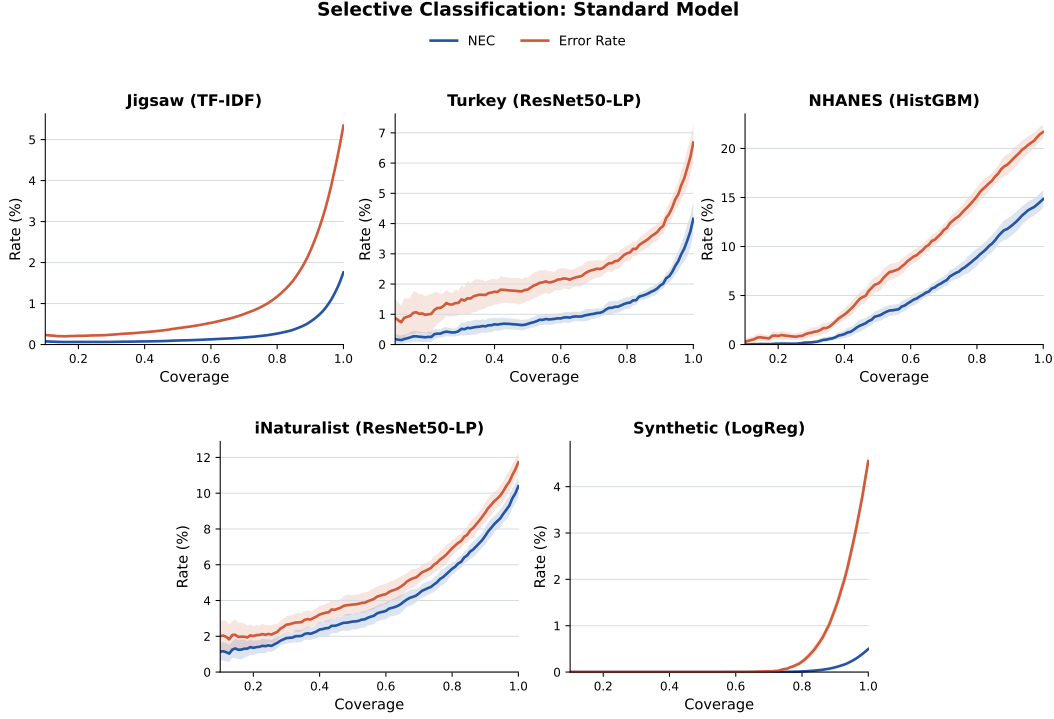


Figure 6. Selective-classification curves for the Standard models. At each coverage, NEC and Error Rate are computed on the same confidence-ranked retained subset. Curves are means over ten seeds; shaded bands are 95% confidence intervals.

Across all five datasets, both metrics are lower for the most-confident 10% than at full coverage, and mean NEC is no greater than mean Error Rate at any plotted coverage. The curves need not be pointwise monotone because newly included examples can differ in both error frequency and cost composition.

This selective coverage analysis complements the full-coverage evaluation reported in Section 4.2.